MECH191 — Metallurgy

Week 16

Corrosion & Course Wrap-up

December 7, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Corrosion Fundamentals & Types

The electrochemical cell, uniform, galvanic, pitting, crevice, and stress corrosion cracking — mechanisms and recognition

02

Protection, Material Selection & Inspection

Coatings, cathodic protection, inhibitors, design decisions, and corrosion monitoring methods

03

Course Wrap-up

How the five units connect — from atomic structure to failure analysis — and what it means to be a metallurgy-literate technician

Alloy Research Portfolio — Module 15

No Microscope Required

Module 15 • Week 16

Course Capstone & Synthesis

What to do this week:

No new research this week — synthesize your notes from Modules 1–14 into the final capstone questions.

Steps:

1 Research — record data & sources
2 Answer — short-answer questions
3 Apply — critical-thinking & equations

Alloys of Record

· No new alloy — synthesis exercise

How to Complete This Module:

1 Research the alloy(s) using credible sources
2 Record data with a source citation
3 Answer the short-answer questions
4 Answer the critical-thinking questions
5 Keep the module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16 ← here** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM • December 16, 2026

Week 16 — At a Glance

Topics

Electrochemical corrosion — the four-element cell, galvanic series, and passivation
Corrosion types — uniform, galvanic (area effect), pitting (PREN), crevice, SCC, intergranular
Coatings — paint systems, surface preparation (Sa 2.5), hot-dip galvanizing, thermal spray, anodizing
Cathodic protection — sacrificial anodes vs. impressed current, -0.85V criterion per NACE SP0169
Design for corrosion — eliminate crevices, ensure drainage, isolate dissimilar metals, control residual stress
Inspection & monitoring — UT thickness, pulsed eddy current for CUI, ER probes, corrosion coupons
Course wrap-up — connecting the five units into a single integrated framework

Resources

Reference Material

Moniz Ch. 31
Corrosion

Reference Pages

Stainless Steel PREN — Grade Comparison
Corrosion Protection Guide — NACE / ISO 12944
ASM Vol. 13A — Corrosion Fundamentals

Standards

NACE SP0169 — CP Criteria
ISO 8501 — Surface Preparation
NACE MR0175 — H_2S Service

Final Exam · December 16, 2026

5:00 – 7:00 PM · Same location as class

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What is corrosion and why is it thermodynamically driven?
2. What are the main types of corrosion and how is each recognized?
3. What protection methods are available and how do you choose between them?
4. How does material selection and design influence corrosion risk?

Electrochemical Corrosion — The Cell & Passivation

The Four-Element Corrosion Cell

All four elements must be present simultaneously for electrochemical corrosion to occur. Remove any one and the corrosion cell stops — this is the basis of every protection strategy.

Anode (oxidation)

Metal atoms lose electrons and dissolve as ions: $M \rightarrow M^{2+} + 2e^-$. This is where material loss occurs. The anode is the more active (less noble) metal or region.

Cathode (reduction)

Electrons arriving from the anode drive reduction. Alkaline/neutral: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Acidic: $2H^+ + 2e^- \rightarrow H_2$. No material loss at the cathode.

Electrolyte

Conducting liquid medium — water with dissolved salts, acids, or gases. Conductivity increases with salt concentration and temperature. Seawater is highly corrosive because of its high conductivity.

Metallic path

Electrical connection between anode and cathode — can be the metal itself (different microstructural features or two metals in contact) or an external conductor.

Galvanic Series & Passivation

Galvanic series

Metals ranked by electrochemical potential in seawater. More active (less noble) metals are anodic and corrode preferentially when coupled to more noble metals. Greater separation in the series = more severe galvanic attack.

Active (anodic) end

Magnesium → Zinc → Aluminum → Carbon steel

Mid-range

Cast iron → Lead → Tin → Copper alloys → Bronzes

Noble (cathodic) end

Stainless steel (passive) → Titanium → Gold → Platinum

Passivation

Some metals form a stable, adherent oxide that dramatically slows further corrosion:

Stainless steel (Cr_2O_3) · Aluminum (Al_2O_3) · Titanium (TiO_2)

Passivity destroyed by: chloride ions, mechanical damage, or reducing conditions. Once broken, the metal corrodes actively until the passive film can reform — if ever.

Corrosion Types — Uniform & Galvanic

Uniform (General) Corrosion

Mechanism

Anodic and cathodic reactions distributed uniformly across the surface — no localization. The most predictable form of corrosion because the rate is measurable and stable.

Recognition

Even surface loss, uniform scale or rust layer. Corrosion rate measured in mm/year or mils per year (MPY) — allows remaining life calculations.

Design approach

Specify a material with a known corrosion rate for the environment, then add a corrosion allowance to the wall thickness. Monitor with UT thickness measurement at regular intervals throughout service life.

Examples

Rusting of unprotected carbon steel in atmosphere or soil. Acid attack on steel in pickling baths. Uniform thinning of tank shells in process environments.

Galvanic Corrosion

Mechanism

Two dissimilar metals in electrical contact in an electrolyte — potential difference drives current; the less noble metal (anode) corrodes at an accelerated rate while the more noble metal (cathode) is protected.

Area effect — critical

Large cathode + small anode = severe attack; large anode + small cathode = mild. Zinc on steel works: Zn is the large anode, exposed steel at scratches is a tiny cathode. Never coat the anode only — any break creates a concentrated small-anode attack.

Recognition

Concentrated corrosion on the less noble material, particularly near the junction between the two metals. The more noble material shows little or no attack.

Prevention & Examples

Electrical isolation (non-conducting gaskets, sleeves); select metals close in galvanic series; apply protective coating to the cathode. Examples: zinc anodes on ship hulls; Al fittings on SS pipework; carbon steel fasteners in copper alloy corroding rapidly.

Corrosion Types — Pitting, Crevice & Stress Corrosion Cracking

Pitting & Crevice Corrosion

Pitting Corrosion

Local passive film breakdown → small anode surrounded by large passive cathode → pit becomes self-sustaining. Metal ions accumulate inside the pit, attracting Cl^-, dropping pH, preventing repassivation. Recognition: discrete pits on otherwise intact surface — depth concealed by thin corrosion product.

PREN = %Cr + 3.3×%Mo + 16×%N → higher = better pitting resistance

304 (PREN ~19) · 316 (PREN ~24) · Duplex 2205 (PREN ~35–40)

Crevice Corrosion

Stagnant electrolyte in a tight gap (0.025–0.1 mm) depletes oxygen → interior becomes anodic → metal ions accumulate, pH drops, Cl^- migrates in → aggressive chemistry prevents repassivation. Recognition: damage concentrated under gaskets, lap joints, bolt heads, and deposits.

Prevention: butt joints over lap joints; solid/PTFE-filled gaskets; continuous rather than intermittent welds; higher PREN alloys; ensure drainage.

Stress Corrosion Cracking (SCC)

Mechanism

Tensile stress (residual or applied) + specific corrosive environment = cracking at stresses far below yield strength. Involves anodic dissolution at crack tip and hydrogen embrittlement. Crack propagates stably until the remaining cross-section fractures by overload.

Critical material-environment pairs

Austenitic SS + chlorides. High-strength steel + H_2S (sulfide stress cracking — NACE MR0175). Brass + ammonia (season cracking). Aluminum alloys + chlorides. Titanium + methanol or red fuming nitric acid.

Recognition

Branching, intergranular or transgranular crack network. Minimal plastic deformation. Brittle fracture surface with no beach marks — this is the key distinction from fatigue. Branching is the most characteristic macroscopic feature visible in the field.

Prevention & Detection

Stress relief (PWHT), shot peening for compressive RS. Select resistant alloys (duplex SS for chlorides, low-alloy steel + PWHT for H_2S). Modify environment (remove Cl^- or O_2; inhibitors; maintain pH >7). Detection: PT, ACFM through coatings, UT for subsurface.

Corrosion Protection — Coatings & Cathodic Protection

Protective Coatings

Organic paint systems

Three-coat system: zinc-rich primer (sacrificial — zinc corrodes to protect steel even when scratched) + intermediate barrier coat + UV/abrasion-resistant topcoat. Surface preparation is the single most important factor — blasting to Sa 2.5 per ISO 8501. Holiday testing detects pinholes before commissioning.

Hot-dip galvanizing

Steel immersed in molten zinc at ~450°C — zinc metallurgically bonds through a series of Fe-Zn alloy layers. Provides both barrier and sacrificial protection. Zinc corrodes preferentially at scratches and cut edges. Service life 20–50 years in mild environments.

Thermal spray

Zinc or aluminum powder/wire melted and propelled onto the surface — applied to large structures in situ (bridges, offshore structures). Thicker than electroplating; can be applied without immersion.

Anodizing & conversion coatings

Anodizing thickens the oxide on aluminum electrochemically — improves corrosion and wear resistance. Phosphating provides paint-adhesion base for automotive steel. Chromate conversion being phased out (toxicity); trivalent alternatives increasingly specified.

Cathodic Protection & Inhibitors

Cathodic protection (CP)

Makes the structure the cathode of an electrochemical cell — corrosion only occurs at anodes.

Sacrificial anode CP

Zn, Mg, or Al anode connected to the structure — the anode corrodes preferentially. Passive, no power required, self-regulating. Ship hulls, buried pipelines, tank floors. Anodes must be inspected and replaced periodically.

Impressed current CP (ICCP)

DC power source drives current from inert anode (graphite, MMO) onto the structure. Controllable and suitable for large installations. Buried/submerged pipelines, storage tanks, reinforced concrete. Requires monitoring — power supply failure leaves structure unprotected.

CP criterion (NACE SP0169): −0.85 V vs. $Cu/CuSO_4$ reference electrode

Inhibitors

Anodic (form anode film — risk if under-dosed) · Cathodic (reduce cathodic rate — safer under-dosed) · Mixed (act on both — cooling water, pipelines, preservation fluids).

Design, Material Selection & Corrosion Inspection

Design & Material Selection

Material selection is the most permanent defense — an inherently resistant alloy requires no maintenance and won't fail if the protection system is damaged.

Joint design

Butt joints preferred over lap joints — lap joints create crevices. Continuous welds instead of intermittent welds seal gaps. Full-penetration seal welds at tube-to-tubesheet joints eliminate the crevice between tube OD and bore.

Drainage & surface finish

Slope surfaces so water drains — horizontal surfaces that pool water corrode faster. Smooth surfaces are more corrosion resistant than rough (rough surfaces trap moisture and contamination). Smooth machined keyways + polished fillets reduce both corrosion and fatigue initiation.

Dissimilar metal isolation

Electrically isolate dissimilar metals with non-conducting gaskets, sleeves, and washers. If isolation is impractical, select metals close together in the galvanic series or ensure the less noble material has a much larger area than the more noble.

Residual stress control

Tensile residual stress from welding or machining accelerates SCC. PWHT for stress relief. Shot peening introduces compressive residual stress at the surface — a common SCC and fatigue prevention measure for welded joints and keyways.

Corrosion Inspection & Monitoring

Visual inspection

First step — rust staining, discoloration, paint blistering, pitting, deposits, and scaling. Requires adequate lighting, access, and mirrors or borescopes for confined spaces.

UT thickness measurement

Primary method for measuring remaining wall thickness in corroded piping, vessels, and structural members — from one side without internal access. Results compared to minimum acceptable thickness (fitness for service).

Corrosion Under Insulation (CUI)

Corrosion hidden beneath insulation — cannot be seen without stripping. Pulsed eddy current (PEC) and radiographic profile techniques screen through insulation without removal. A significant inspection challenge in process plant.

Monitoring — rate measurement

ER probes and LPR probes: real-time corrosion rate in process pipelines and vessels. Corrosion coupons: pre-weighed metal samples exposed for a defined period — direct weight loss measurement. Used to verify inhibitor programs and detect changes in process chemistry.

Course Wrap-Up — How the Five Units Connect

These are not five separate subjects. They form a single chain — each unit explains why the next one matters.

Unit 1 — Materials Fundamentals Aug 24 – Sep 7	Atomic bonding explains conductivity and electrochemical corrosion. Crystal structure determines slip systems, deformation behavior, and SCC crack paths. Grain boundaries are sites of sensitization, intergranular corrosion, and hydrogen trapping.
Unit 2 — Steel & Alloys Sep 14 – Sep 28	Carbon equivalent drives preheat requirement and determines HIC susceptibility. Stainless grade determines pitting (PREN) and SCC resistance. Heat treatment condition sets HAZ hardenability — the microstructure that either resists or fails in service.
Unit 3 — Testing, Inspection & Quality Oct 5 – Oct 19	NDT methods detect the defects that Unit 2's microstructure makes and Unit 4's processes create. Hardness testing verifies heat treatment. Inspection documentation creates the traceability that makes a failure investigation possible.
Unit 4 — Manufacturing Processes Oct 26 – Nov 16	Casting defects provide corrosion and fatigue initiation sites. Forming introduces residual stress that drives SCC. Machining quality determines surface finish and residual stress state — the foundation for fatigue life in service.
Unit 5 — Joining, Failure & Corrosion Nov 23 – Dec 7	The weld HAZ is a microstructure gradient each zone with different corrosion resistance. Residual weld stress drives SCC. Failure analysis connects all previous units — explaining why a component failed and feeding that knowledge back into design, material selection, and inspection.

Technician's Perspective

Corrosion type identification in the field — what the technician sees and what it means.

Red Flag: Leak at 304 SS tube-to-tubesheet joint in seawater heat exchanger — 14 months in service

Inspection reveals three overlapping corrosion mechanisms: (1) Galvanic — carbon steel tubesheet is anodic to stainless tubes; unfavorable area ratio (small anode/large cathode) drives accelerated attack at the tubesheet. (2) Crevice — the tight gap between tube OD and tubesheet bore in stagnant seawater depletes O_2, drops pH, concentrates Cl^- — classic crevice geometry. (3) Sensitization — if standard 304 (not 304L) was used without interpass temperature control, chromium carbide precipitation in the HAZ depletes Cr below the passivity threshold. Action: specify 304L or duplex 2205 for seawater; specify full-strength seal welds to eliminate the crevice; apply sacrificial zinc anodes to the carbon steel tubesheet; add to failure investigation as a multi-mechanism case study.

Caution: Branching cracks at weld toes on austenitic stainless pressure vessel in chloride service

VT finds tight, branching cracks running from weld toes. No beach marks on the fracture surface (distinguishing SCC from fatigue). No plastic deformation. Metallographic sections show transgranular cracking with fine branching — the textbook appearance of chloride SCC in austenitic stainless. Root cause: tensile weld residual stress + chloride from insulation + elevated temperature. Action: PWHT for stress relief (not always possible on austenitic SS — sensitization risk); shot peen weld toes; review insulation specification (chloride-free insulation required per NACE for SS service); consider upgrade to duplex stainless or super-austenitic for remaining vessels in the same service.

Good Practice: Holiday testing a coating before cathodic protection commissioning — finding the pinhole that prevents a concentrated attack

A newly coated pipeline section is holiday tested before backfilling using a high-voltage spark tester. A pinhole is found at a fitting where the coating was not fully sealed. The pinhole is repaired and retested. Without the repair, the pinhole under CP would have created a small anode/large cathodically protected cathode situation — concentrating the full galvanic driving force at a single point and producing rapid localized attack. Action: holiday testing is a mandatory step before burying or commissioning any cathodically protected coated structure; document all findings and repairs; retest after each repair until the coating is holiday-free.

Core Question Answers

What is corrosion and why is it thermodynamically driven?

Corrosion is the degradation of a metal through chemical or electrochemical reaction with its environment. It is thermodynamically driven because most metals exist naturally as ores (lower energy state) — refining requires energy input, and corrosion is the metal releasing that energy and returning toward its natural state. Electrochemical corrosion requires four elements simultaneously: anode, cathode, electrolyte, and metallic path. Remove any one element and corrosion stops.

What are the main types of corrosion and how is each recognized?

Uniform: even surface loss, predictable rate. Galvanic: concentrated attack on the less noble metal at a dissimilar metal junction. Pitting: discrete pits on otherwise intact surface — depth concealed by corrosion product; driven by local passive film breakdown and Cl^- concentration; PREN quantifies resistance. Crevice: damage under gaskets, lap joints, deposits — stagnant O_2-depleted chemistry. SCC: branching, intergranular/transgranular cracks with no beach marks, no plastic deformation, driven by tensile stress + specific environment.

What protection methods are available and how do you choose between them?

Coatings (barrier and sacrificial) + surface prep is #1 factor. Hot-dip galvanizing for structural steel: barrier + sacrificial, 20–50 yr life. Cathodic protection: sacrificial anodes (passive, ships/pipelines) or ICCP (controllable, large structures). Inhibitors: anodic (risk if under-dosed), cathodic (safer). Material selection: most permanent — match PREN, CE, and alloy to the specific environment. Design: eliminate crevices, ensure drainage, isolate dissimilar metals, control residual stress. Most effective systems combine multiple methods.

How does material selection and design influence corrosion risk?

Material selection must match the specific corrosion mechanism — 304 SS resists uniform attack but fails in Cl^- pitting; copper is excellent in freshwater but fails in ammonia. PREN guides stainless selection for chloride service. CE and alloy composition determine weldability and HAZ hardenability. Design decisions: butt joints over lap joints (no crevice); sloped surfaces for drainage; non-conducting isolation for dissimilar metals; smooth surface finish; PWHT and shot peening to control residual stress and prevent SCC.

Key Takeaways — Week 16: Corrosion & Course Wrap-up

1

Corrosion is thermodynamically inevitable — the goal is not to prevent it absolutely but to understand the specific mechanism operating in a given material-environment combination, slow the rate to an acceptable level, detect it early, and design structures with enough allowance and protection to achieve the required service life.

2

Every corrosion type has a characteristic appearance that allows a technician to identify it in the field — uniform surface loss, galvanic concentration at dissimilar metal junctions, discrete pitting, crevice location under gaskets and deposits, SCC branching — recognition is the first step toward prevention and remediation.

3

The five units of this course are not separate subjects but a single interconnected body of knowledge — atomic structure determines properties, properties determine material selection, processing determines microstructure, microstructure determines performance, and inspection and failure analysis close the loop; a technician who understands these connections is not just a procedure-follower but a problem-solver.

Resources & Final Exam

Week 16 Resources

Moniz Ch. 31
Corrosion and Its Prevention — primary reading

Stainless Steel PREN — Grade Comparison Table
Service environment guide for stainless alloy selection

Corrosion Protection Guide — NACE / ISO 12944 / ASM 13A
Coating selection, surface preparation, and cathodic protection

NACE SP0169
Standard for cathodic protection of buried pipelines — CP criteria

NACE MR0175 / ISO 15156
Material requirements for H_2S service (sour service)

ASM Vol. 13A — Corrosion Fundamentals
Primary reference for corrosion mechanisms and materials selection

Final Exam

Week 17 • December 16, 2026

5:00 PM – 7:00 PM · Same classroom

Exam covers all 5 units:

- Unit 1 — Atomic structure, bonding, mechanical properties
- Unit 2 — Steel families, heat treatment, phase diagrams
- Unit 3 — NDT methods, mechanical testing, quality standards
- Unit 4 — Casting, forming, machining, powder metallurgy
- Unit 5 — Welding, weld metallurgy, fracture, corrosion

Exams must be returned by 7:00 PM to be graded.